Chemical equilibrium

The state of a reversible reaction when the two opposing reactions occur at the same rate and the concentrations of reactants and products do not change with time.

Furthermore, the true equilibrium of a reaction can be attained from both sides. Thus the equilibrium concentrations of the reactants and products are the same whether we start with A and B, or C and D.

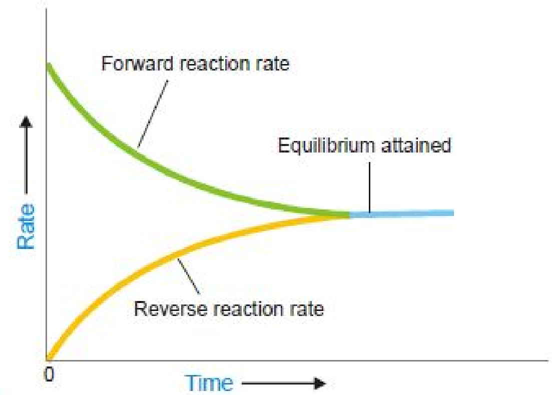

Figure 17.1
At equilibrium the forward reaction rate equals the reverse reaction rate.

CHARACTERISTICS OF CHEMICAL EQUILIBRIUM

(1) Constancy of concentrations
(2) Equilibrium can be initiated from either side
(3) Equilibrium cannot be Attained in an Open Vessel
(4) A catalyst cannot change the equilibrium point
(5) Value of Equilibrium Constant does not depend upon the initial concentration of reactants
(6) At Equilibrium $\Delta G = 0$

LAW OF MASS ACTION

The rate of a chemical reaction is proportional to the active masses of the reactants. Or the rate of a reaction is proportional to the molar concentrations of the reactants.

EQUILIBRIUM CONSTANT: EQUILIBRIUM LAW

The product of the equilibrium concentrations of the products divided by the product of the equilibrium concentrations of the reactants, with each concentration term raised to a power equal to the coefficient of the substance in the balanced equation.

EQUILIBRIUM CONSTANT EXPRESSION IN TERMS OF PARTIAL PRESSURES

When all the reactants and products are gases, we can also formulate the equilibrium constant expression in terms of partial pressure. The relationship between the partial pressure (p) of any one gas in the equilibrium mixture and the molar concentration follows from the general ideal gas equation

$$pV = nRT \quad \text{or} \quad p = \left(\frac{n}{V}\right) RT$$

The quantity $\frac{n}{V}$ is the number of moles of the gas per unit volume and is simply the molar concentration. Thus,

$$p = (\text{Molar concentration}) \times RT$$

i.e., **the partial pressure of a gas in the equilibrium mixture is directly proportional to its molar concentration at a given temperature.** Therefore, we can write the equilibrium constant expression in terms of partial pressure instead of molar concentrations. For a general reaction

$$l\mathbf{L}(g) + m\mathrm{M}(g) \rightleftharpoons y\mathrm{Y}(g) + z\mathrm{Z}(g)$$

the equilibrium law or the equilibrium constant may be written as

$$K_p = \frac{(p_\mathrm{Y})^y (p_\mathrm{Z})^z}{(p_\mathrm{L})^l (p_\mathrm{M})^m}$$

Here K_p is the equilibrium constant, the subscript p referring to partial pressure. Partial pressures are expressed in atmospheres.

Relation between k_p and k_c

Let us consider a general equation

$$j\mathrm{A} + k\mathrm{B} \Longleftrightarrow l\mathrm{C} + m\mathrm{D}$$

when all reactants and products are gases. We can write the equilibrium constant expression in terms of partial pressures as

$$K_p = \frac{(p_C)^l (p_D)^m}{(p_A)^j (p_B)^k} \qquad ...(1)$$

Assuming that all these gases constituting the equilibrium mixture obey the ideal gas equation, the partial pressure (p) of a gas is

$$p = \left(\frac{n}{V}\right) RT$$

Where $\frac{n}{V}$ is the molar concentration. Thus the partial pressures of individual gases, A, B, C and D are:

$$P_\mathrm{A} = [\mathrm{A}]RT;\ p_\mathrm{B} = [\mathrm{B}]RT;\ p_C = [\mathrm{C}]RT;\ p_\mathrm{D} = [\mathrm{D}]RT$$

Substituting this value in equation 1, we have

$$K_p = \frac{[\mathrm{C}]^l (RT)^l [\mathrm{D}]^m (RT)^m}{[\mathrm{A}]^j (RT)^j [\mathrm{B}]^k (RT)^k}$$

or

$$K_p = \frac{[\mathrm{C}]^l [\mathrm{D}]^m}{[\mathrm{A}]^j [\mathrm{B}]^k} \times \frac{(RT)^{l+m}}{(RT)^{j+k}}$$

$$K_p = K_c \times (RT)^{(l+m)-(j+k)}$$

$\therefore \qquad K_p = K_c \times (RT)^{\Delta n} \qquad$ ...(2)

where $\Delta n - (l + m) \quad (j + k)$, the difference in the sums of the coefficients for the gaseous products and reactants.

From the expression (2) it is clear that when $\Delta n = 0$, $K_p = K_c$.

SOLVED PROBLEM 1. At 500°C, the reaction between N_2 and H_2 to form ammonia has $K_c = 6.0 \times 10^{-2}$. What is the numerical value of K_p for the reaction?

SOLUTION

Here, we will use the general expression

$$K_p = K_c (RT)^{\Delta n}$$

For the reaction $\quad N_2 + 3H_2 \rightleftharpoons 2NH_3$

we have $\quad \Delta n$ = (sum of quctients of products) – (sum of quotients of reactants)

$$= 2 - 4 = -2$$

$$K_c = 6.0 \times 10^{-2}$$

$$T = 500 + 273 = 773 \text{ K}$$

$$R = 0.0821$$

Substituting the value of R, T, K_c and Δn in the general expression, we have

$$K_p = (6.0 \times 10^{-2}) [(0.0821) \times (773)]^{-2}$$

$$= \mathbf{1.5 \times 10^{-5}}$$

SOLVED PROBLEM 2. The value of K_p at 25°C for the reaction

$$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$$

is 1.9×10^3 atm^{-1}. Calculate the value cf K_c at the same temperature.

SOLUTION

We can write the general expression as

$$K_p = K_c (RT)^{\Delta n} \text{ cr } K_c = \frac{K_p}{(RT)^{\Delta n}}$$

Here,

$$T = 25 + 273 = 298 \text{ K}$$

$$R = 0.0821$$

$$\Delta n = 2 - (2 + 1) = -1$$

$$K_p = 1.9 \times 10^3$$

Substituting these values in the general expression

$$K_c = \frac{1.9 \times 10^3}{(0.0821 \times 298)^{-1}}$$

$$= \mathbf{4.6 \times 10^4}$$

SOLVED PROBLEM 2. Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C. When equilibrium is established, 3.00 moles of N_2, 2.10 moles of H_2 and 0.298 mole of NH_3 are found to be present. Find the value of K_c for the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

at 500°C.

SOLUTION

The equilibrium concentrations are obtained by dividing the number of moles of each reactant and product by the volume, 5.00 litres. Thus,

$$[N_2] = 3.00 \text{ mole}/5.00 \text{ L} = 0.600 \text{ M}$$
$$[H_2] = 2.10 \text{ mole}/5.00 \text{ L} = 0.420 \text{ M}$$
$$[NH_3] = 0.298 \text{ mole}/5.00 \text{ L} = 0.0596 \text{ M}$$

Substituting these concentrations (not number of moles) in the equilibrium constant expression, we get the value of K_c.

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.0596)^2}{(0.600)(0.420)^3} = 0.080$$

Thus, for the reaction of H_2 and N_2 to form NH_3 at 500°C, we can write

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \mathbf{0.080}$$

SOLVED PROBLEM 4. At a certain temperature, 0.100 mole of H_2 and 0.100 mole of I_2 were placed in a one-litre flask. The purple colour of iodine vapour was used to monitor the reaction. After a time, the equilibrium

$$H_2 + I_2 \rightleftharpoons 2HI$$

was established and it was found that the concentration of I_2 decreased to 0.020 mole/litre. Calculate the value of K_c for the reaction at the given temperature.

SOLUTION

The equilibrium constant expression for the reaction

$$H_2 + I_2 \rightleftharpoons 2HI$$

is

$$K_c = \frac{[HI]^2}{[H_2][I_2]}$$

Let us proceed to find the equilibrium concentrations of the various species.

The initial moles of H_2, I_2 and HI present at equilibrium are

$$\begin{array}{lccccc} & H_2 & + & I_2 & \rightleftharpoons & 2HI \\ \text{moles at equilibrium :} & (0.100-x) & & (0.100-x) & & 2x \end{array}$$

Since the volume of the reaction vessel is one litre, the moles at equilibrium also represent the equilibrium concentrations. Thus,

$$[I_2] = 0.100 - x = 0.020 \text{ (given)}$$

or

$$x = 0.100 - 0.020$$
$$= 0.080$$

Knowing the value of x, we can give the equilibrium concentrations of H_2, I_2 and HI

$$[H_2] = 0.100 - x = 0.100 - 0.080 = 0.020$$
$$[I_2] = 0.020 \text{ (given)}$$
$$[HI] = 2x = 0.160$$

Substituting these values in the equilibrium constant expression, we get the value of K_c.

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(0.160)^2}{0.02 \times 0.02} = \mathbf{64}$$

LE CHATELIER'S PRINCIPLE

If a change in concentration, pressure or temperature is caused to a chemical reaction in equilibrium, the equilibrium will shift to the right or the left so as to minimize the change.

$N_2 + 3H_2 \longleftrightarrow 2NH_3 + 22.2$ kcal

EFFECT OF CHANGE IN CONCENTRATION

We can restate Le Chatelier's principle for the special case of concentration changes: when concentration of any of the reactants or products is changed, the equilibrium shifts in a direction so as to reduce the change in concentration that was made.

Effect of change of concentration on Ammonia Synthesis reaction Let us illustrate the effect of change of concentration on a system at equilibrium by taking example of the ammonia synthesis reaction:

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

When N_2 (or H_2) is added to the equilibrium already in existence, the equilibrium will shift to the right so as to reduce the concentration of N_2 (Le Chatelier's principle).

EFFECT OF A CHANGE IN PRESSURE

When pressure is increased on a gaseous equilibrium reaction, the equilibrium will shift in a direction which tends to decrease the pressure.
Let us consider a reaction,

$A + B \rightleftarrows C$

The combination of A and B produces a decrease of number of molecules while the decomposition of C into A and B results in the increase of molecules. Therefore, by the increase of pressure on the equilibrium it will shift to right and give more C. A decrease in pressure will cause the opposite effect. The equilibrium will shift to the left when C will decompose to form more of A and B.

EFFECT OF CHANGE OF TEMPERATURE

When temperature of a reaction is increased, the equilibrium shifts in a direction in which heat is absorbed.
Let us consider an exothermic reaction

$A + B \longleftrightarrow C + \text{heat}$

When the temperature of the system is increased, heat is supplied to it from outside. According to Le Chatelier's principle, the equilibrium will shift to the left which involves the absorption of heat. This would result in the increase of the concentration of the reactants A and B.